

HW3ROME_report_zw4892

PART1:

1. iteration0

keeps give out "warning compiling testbench"

find out in the function "get_module" will return with the special symbol like "{,]"

so we have to clean the return code.

2. Iteration1

Testbench may not match when generate the testbench. Like the port names are (in1,in2) but ports in the testbench which generated are (a,b). So we have to change the code in the submodule.

3. Iteration2

The code can pass the logic mux2to1 but after it the logic of mux4to1 and mux8to1 can't pass the test. Because in the function of write_code_blocks_to_file ,the function always capture the first module AI wrote. So change the capture function and let it get all the modules.

4. Iteration3

The testbench wasn't put in the correct folder so the code can't search out the correct testbench. And change the prompts to let AI "generate module by using the two modules mux4to1 and one module mux2to1"

5. Iteration4

It turned out that it pass all the test.

```

*** File ./mux2to1/mux2to1.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! mux2to1 logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./mux4to1/mux4to1.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! mux4to1 logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./mux8to1/mux8to1.v has been cleaned and written.
    compile failed: ./mux8to1tb.v:6: error: Unknown module type: mux8to1
2 error(s) during elaboration.
*** These modules were missing:
    mux8to1 referenced 1 times.
***

File ./mux8to1/mux8to1.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! mux8to1 logic pass!
Total time: 8.784162282943726
Generation time: 8.777100563049316
Error handling time: 0.007061481475830078
Overall Total time: 8.784162282943726
Overall Generation Time: 8.777100563049316
Overall Error handling time: 0.007061481475830078

```

PART2:

Iteration0:

Didn't use the modules before to generate the whole adder module so change the prompts.

```

hier_gen(submodules)
== File ./half_adder/half_adder.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! half_adder logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./full_adder/full_adder.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! full_adder logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./adder/adder.v has been cleaned and written.
    compile failed: ./adder4tb.v:7: error: Unknown module type: adder4
3 error(s) during elaboration.
== These modules were missing:
    adder4 referenced 1 times.
==

KeyboardInterrupt: Traceback (most recent call last):
  File ./adder4tb.v:7: error: Unknown module type: adder4
    hier_gen(submodules)

22 frames
./usr/lib/python3.12/site-packages/verilog/verilog.py in read(self, line, buffer)
1103         return self._obj.read(line, buffer)
1104     else:
1105         return self._obj.read(line)
-> 1106     except OSError as e:
1107         if e.args[0] == EPIPE and self._suppress_ragged_eds:
KeyboardInterrupt:

```

Iteration1:

The port name didn't match with the name in testbench so change the testbench's port names.

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Iteration2:

passed

```
*** File ./half_adder/half_adder.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! half_adder logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./full_adder/full_adder.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! full_adder logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
File ./adder4/adder4.v has been cleaned and written.
    compile success, start simulation...
    simulation outputs: passed!

    congrats! adder4 logic pass!
Total time: 0.0
Generation time: 0.0
Error handling time: 0.0
Overall Total time: 0.0
Overall Generation Time: 0.0
Overall Error handling time: 0.0
```